RESEARCH PROPOSAL | PRE-SPECIFIED PROTOCOL

A Pre-Specified Comparison of 2D, 2.5D and Lightweight 3D U-Nets on ATM'22

Airway completeness, neighbouring-slice context and computational cost in pulmonary airway tree segmentation

Author	Muhammad Mubeen
Advisor	Dr Amir Khan
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Nature of this study

This is a controlled, reproducible measurement, not a new method. No new architecture is proposed. Results will be reported regardless of which configuration performs best.

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Executive Summary

Research question	How much does three-slice adjacent axial context improve airway completeness over single-slice 2D, and how does a practical 2.5D U-Net compare with a lightweight 3D U-Net on ATM'22?
Design	Pre-specified, five-fold cross-validated comparison of three U-Net configurations under a shared training protocol.
Models	• Single-slice 2D U-Net • Three-slice 2.5D U-Net • Lightweight patch-based 3D U-Net
Primary outcomes	Branch detection rate (BD) and tree length detected rate (TD), per scan.
Secondary outcomes	Dice, precision, approximate diameter-stratified recall, fragmentation, and computational cost (training memory, training time, inference latency, inference memory).
Primary comparisons	• **P1:** 2.5D minus 2D (superiority) • **P2:** 2.5D minus 3D (estimation; conditional non-inferiority only if a defensible margin is approved)
Statistical approach	Fold-stratified paired bootstrap, 10,000 resamples, inference conditional on fitted cross-validation models.
Data	ATM'22: ~299 usable labelled training scans; 30 development scans; 269 scans in five fixed held-out folds.
Compute	One fixed GPU; target 30 training jobs (9 pilot + 15 core + 6 conditional seed-sensitivity).
Timeline	32 weeks: registration and review (1–3), data preparation (4–7), pilot and freeze (8–11), main experiment (12–21), analysis (22–27), writing and release (28–32).
Contribution	A controlled, reproducible, pre-specified measurement of the incremental effect of neighbouring-slice context on airway completeness and its relationship to computational cost. No new architecture is proposed.
Not claimed	Novelty of 2.5D segmentation, state-of-the-art performance, universal efficiency, clinical effectiveness, or generalisation beyond the benchmark.

1. Abstract

Problem. Pulmonary airway tree segmentation from chest computed tomography (CT) supports airway morphology analysis, respiratory research and planning for bronchoscopic procedures. Peripheral airways are narrow, low in contrast and form a complex branching structure, so a segmentation can achieve a high volumetric Dice score while missing substantial portions of the peripheral tree. Airway-specific completeness measures are therefore important.

Design. This study conducts a pre-specified comparison of three U-Net configurations on the ATM'22 benchmark: a single-slice 2D U-Net, a three-slice adjacent-slice 2.5D U-Net and a lightweight patch-based 3D U-Net.

Outcomes. The primary outcomes are branch detection rate (BD) and tree length detected rate (TD), evaluated per scan. Dice, precision, approximate diameter-stratified reference recall and prediction fragmentation are secondary. Training memory, training time, inference latency and inference memory are measured on fixed hardware.

Analysis. The principal comparison estimates the incremental benefit of adjacent axial-slice context (2.5D versus 2D within the same architectural family). The 2.5D versus 3D comparison estimates the difference between two practical configurations, not dimensionality alone. Differences are assessed with paired, fold-stratified bootstrap intervals, explicitly conditional on the fitted cross-validation models.

Contribution. No new architecture is proposed and no assumption is made that 2.5D will outperform 3D. The contribution is a controlled, reproducible assessment of airway completeness and computational requirements, reported regardless of outcome.

2. Background and Motivation

2.1 Clinical and Technical Motivation

Automated airway segmentation extracts the airway tree from thoracic CT. The output supports airway morphology analysis, computational respiratory research, preoperative planning and bronchoscopic navigation, particularly when peripheral bronchi must be identified beyond the major central airways. The airway tree presents a distinct challenge:

- The trachea and main bronchi are large and generally have clearer boundaries.
- Distal branches may occupy only a few voxels.
- Partial-volume effects, low contrast, noise, disease-related variation and adjacent vessels make small airways hard to separate from surrounding tissue.
- Airways form a connected branching system, so a missed region near a junction can disconnect a peripheral branch from the predicted tree.

Volumetric overlap alone therefore does not describe airway segmentation quality. A method may obtain a relatively high Dice score while missing small branches or substantial portions of the tree.

2.2 Dimensionality and Neighbouring-Slice Context

Approach	Input	Key property
2D segmentation	Single image slice	Modest memory requirements; no explicit through-plane context.
2.5D segmentation	Multiple neighbouring slices as input channels	Limited through-plane context without full 3D volume processing.
3D segmentation	Volumetric patches	Direct volumetric context; memory depends on feature-map dimensions, depth, patch size and batch size.

These approaches involve trade-offs rather than a universally superior dimensionality. The benefit of additional context may depend on anatomical structure, resolution, network design, training data, hardware and the evaluation metric.

2.3 Motivation for a Controlled Comparison

Previous airway-segmentation studies have explored adjacent-slice models, 3D U-Nets, connectivity-aware methods, slice propagation and topology-oriented objectives. They establish that spatial context and structural continuity both matter, but they do not answer every practical question about the incremental value of three adjacent axial slices relative to a single slice in a common U-Net architecture. Results from other organs indicate that 3D networks can be more accurate and faster at inference while requiring more memory; these cannot be assumed to hold for pulmonary airways. A controlled comparison determines:

- Whether a small amount of neighbouring-slice context improves airway completeness over a single-slice model.
- How a practical 2.5D configuration compares with a feasible lightweight 3D alternative.

> **Central motivation:** quantify the relationship between airway completeness, neighbouring-slice context and computational resources without assuming the result in advance.

3. Research Questions, Estimand and Hypotheses

3.1 Estimand and Target Population

- For each held-out CT scan, each model produces one primary prediction using the checkpoint selected on that fold's inner-validation set.
- For each comparison and primary metric, the observed benchmark effect is the equally scan-weighted mean paired difference between two models, in percentage points.
- **P1:** 2.5D minus 2D. **P2:** 2.5D minus lightweight 3D.
- A fold-stratified paired bootstrap estimates case-sampling uncertainty. Intervals are conditional on the fitted cross-validation models and are not presented as capturing variability from retraining across random initialisations or training sets.

3.2 Research Questions

Primary

ID	Question	Type
RQ1	Does a three-slice adjacent-slice 2.5D U-Net improve BD and TD compared with a single-slice 2D U-Net?	Superiority
RQ2	What are the differences in BD and TD between the three-slice 2.5D U-Net and the lightweight patch-based 3D U-Net?	Estimation

A non-inferiority assessment of 2.5D relative to 3D is conducted only if a defensible margin is established and approved before the protocol is frozen.

Secondary

ID	Question
RQ3	How do the three configurations differ in training memory, training time, inference latency and inference memory under the specified hardware and inference settings?
RQ4	Does the difference in recall between 2.5D and 2D differ between the smallest and largest approximate reference-diameter groups?
RQ5	How sensitive are BD and TD to random initialisation on one pre-selected cross-validation fold?

RQ4 is a pre-specified secondary analysis. Other analyses by diameter, slice spacing and scan characteristics remain exploratory.

3.3 Hypotheses

H1: 2.5D superiority over 2D

Statement	Three-slice 2.5D improves airway completeness over single-slice 2D.
Decision rule	For BD and TD separately, support requires the two-sided 97.5% fold-stratified bootstrap confidence interval for the mean paired difference (2.5D minus 2D) to lie entirely above zero.
Interpretation	Using 97.5% intervals for each of the two primary metrics is a conservative adjustment for two metric-specific claims. A general claim of improved completeness is made only if both BD and TD satisfy the criterion. If only one does, the result is reported as partial, metric-specific evidence.

H2: conditional non-inferiority of 2.5D relative to 3D

Statement	2.5D is no worse than the lightweight 3D configuration by more than a pre-specified margin, delta, for both BD and TD.
Planning value	[2 absolute percentage points]. Not an established clinical or methodological threshold. Used only if its practical interpretation and justification are approved before the main experiments.
Decision rule	If tested, supported only if the lower endpoint of the two-sided 95% confidence interval for (2.5D minus 3D) exceeds minus delta for both BD and TD. Both conditions must hold; a favourable outcome on one metric cannot compensate for a failure on the other.
Procedure	Because the claim requires both metric-specific conditions, the analysis uses the corresponding intersection-union decision rule. The statistical procedure is reviewed before the protocol is frozen.
If no margin is approved	RQ2 remains estimation-only. Effect sizes and intervals are still reported.
Reporting commitment	The literature gives no sufficient basis to assume 2.5D will match 3D on airway segmentation. A result favouring 3D, including a substantial advantage, is reported without changing the pre-specified analysis.

4. Literature Review

4.1 Airway Segmentation Using 2.5D Methods

Study	Method	Key finding	Relation to this proposal
Yun et al. (2019)	Triplanar 2.5D CNN using axial, sagittal and coronal neighbours	2.5D information contributes to automated airway extraction across datasets	Different formulation (triplanar, voxel-wise); not a direct baseline.
Carmo et al. (2022)	MEDSeg: three-channel adjacent-slice modified EfficientDet, ATM'22	Dice 90.72 (internal), 95.52 (external), 93.49 (test)	Closest prior work. No shared-backbone comparison; no isolation of the adjacent-slice effect.
Charbonnier et al. (2017)	CNN-based leak detection for airway refinement	Illustrates joint importance of completeness and false-positive control	Focus on leak detection, not dimensionality comparison.

4.2 Airway Segmentation Using 3D Methods

Study	Method	Key finding	Relation to this proposal
Garcia-Uceda et al. (2021)	Low-memory 3D U-Net; five resolution levels, 16 initial feature maps; 252^3 patches; batch size 1	Fitted within 24 GB GPU; under 1 min per scan including post-processing	Defines the 3D comparator. Shows 3D is practical; patch size matters.
Qin et al. (2019)	AirwayNet: voxel-connectivity prediction	Addresses structural continuity	Related work; not a baseline.
Wu et al. (2022)	LTSP: long-term slice propagation	Preserves topology and recovers small bronchi	Related work; not a baseline.

4.3 Direct Dimensionality Comparisons in Other Organs

Avesta et al. (2023) compared 2D, 2.5D and 3D segmentation across capsule networks, U-Nets and nnU-Nets using 3,430 brain MRI volumes.

Aspect	Finding
Accuracy	3D > 2.5D > 2D on Dice
Training convergence	3D approximately 20–40% faster

Aspect	Finding
Inference	3D approximately 30–50% faster
Memory	3D required about 20 times more memory

This demonstrates that increased dimensionality can improve accuracy and speed while imposing substantially greater memory requirements. It does not establish the same relationship for pulmonary airways: brain structures differ from narrow, branching airway trees, and the comparison used MRI rather than chest CT. It motivates measuring accuracy, memory and latency but does not determine the expected outcome here.

4.4 Benchmark Evidence and Dataset Considerations

Resource	Description	Relevance
ATM'22 (Zhang et al., 2023)	500 chest CT scans: 300 training, 50 validation, 150 hidden test; 23 teams; BD and TD reported	Primary dataset. Airway-specific outcomes suit completeness analysis.
AeroPath (Støverud et al., 2024)	27 contrast-enhanced CT scans with challenging pathology	Potential external evaluation. Contrast and pathology introduce domain differences relative to ATM'22.
ATM'26	Separate benchmark with binary airway segmentation and branch-wise anatomical labelling	Not combined with ATM'22. Noted for context only.

The usable ATM'22 training set is generally reported as 299 cases after removal of a mismatched case. This count will be confirmed during dataset acquisition.

4.5 Recent Airway-Segmentation Developments

Study	Contribution	Relevance
Zhao et al. (2025), Skeleton2Mask	Sparse skeleton supervision and label propagation	Topological preservation; not a direct dimensionality comparison.
Yang et al. (2025/2026), progressive curriculum learning	Scale-Enhanced U-Net for continuous airway segmentation	Addresses large/small branch imbalance; recent ATM'22 context.
Yuwen et al. (2026), Cupido Loss	Directional and geometric constraints for airway and artery segmentation	Continued interest in structural priors.
Zhang et al. (2026), FDRAS	Failure diagnosis and post-hoc repair under domain shift	Distinguishes raw segmentation quality from repair.

These developments show that airway continuity, branch recovery and robustness remain active topics. The novelty claim must therefore be checked against recent topology-aware and benchmark studies before the protocol is frozen.

4.6 Synthesis

- Adjacent-slice 2.5D methods have already been used for pulmonary airway segmentation, including on ATM'22.
- Low-memory 3D U-Nets can process large airway-related patches, so 3D should not be treated as inherently impractical.
- Connectivity-aware, propagation-based, skeleton-supervised and topology-oriented approaches show the importance of structural continuity.
- Dimensionality comparisons in other organs report different accuracy, speed and memory trade-offs, but are not directly transferable.
- ATM'22 offers an established benchmark with airway-specific completeness measures; its evaluation and the newer ATM'26 challenge must be understood in their own contexts.

Remaining research question

Not whether 2.5D or 3D airway segmentation is novel in itself, but how much adjacent axial-slice information adds to a single-slice model within a common architecture, how that difference varies with approximate airway diameter, and what computational resources the resulting configurations require.

5. Research Gap, Contribution and Novelty

5.1 Research Gap

Both 2.5D and 3D airway-segmentation methods exist and topology is an important consideration. However, the specific effect of adding three adjacent axial slices to a single-slice U-Net has not been established under the proposed shared training and evaluation protocol.

Provisional gap statement

Among the studies reviewed so far, no study has been identified that compares single-slice 2D, three-slice adjacent-axial 2.5D and lightweight patch-based 3D U-Nets on ATM'22 under a shared U-Net family, loss function and training schedule while reporting paired BD and TD differences and a common computational cost profile.

Status: provisional. The completed literature search must check the ATM'22 challenge report, participant methods, recent airway-segmentation papers and any direct dimensionality comparisons before this claim is used in a manuscript.

5.2 Research Contribution

- A pre-specified comparison of 2D, three-slice 2.5D and lightweight patch-based 3D U-Nets on ATM'22.
- Paired, scan-level estimates of BD and TD differences, with a transparent decision rule and explicit limits on inference.
- A completeness-versus-cost profile: peak training memory, training time, inference latency, inference memory, model size and operation counts where practical.
- A pre-specified comparison of the 2.5D versus 2D recall difference between the smallest and largest approximate reference-diameter groups.
- A reproducible implementation with documented partitions, seeds, evaluation scripts and computational environment, subject to benchmark terms.

5.3 Novelty Framing

The study does not claim that 2.5D airway segmentation, U-Net segmentation, BD, TD or computational profiling is novel individually. The proposed contribution is the controlled measurement of the incremental effect of neighbouring-slice context on airway completeness, characterised by approximate diameter and computational cost.

The 2D versus 2.5D comparison is the principal experimental contrast. The 2.5D versus 3D comparison is a practical reference point, with the explicit limitation that the configurations differ in field of view, receptive field, sampling units and computational requirements. The combination is not automatically novel; its value depends on the completed literature review confirming that this comparison has not already been adequately reported.

5.4 Boundaries

The study will not claim

- That adjacent-slice 2.5D segmentation is a new method.
- State-of-the-art performance on ATM'22.
- That 2.5D is faster or more memory-efficient in every setting.
- That the lightweight 3D model represents all 3D airway-segmentation methods.
- Clinical effectiveness, improved patient outcomes or clinical readiness.
- Generalisation to all CT scanners, diseases, populations or annotation protocols.

6. Dataset and Experimental Splits

6.1 Dataset

Benchmark	ATM'22 pulmonary airway tree segmentation benchmark
Original allocation	500 CT scans: 300 training, 50 validation, 150 hidden test
Usable labelled training set	~299 scans (one mismatched case removed)
Planning assumption	299 usable scans; to be confirmed after registration and download
Validation and test use	Challenge validation and hidden test scans are not used for the primary cross-validation analysis because local ground-truth availability is not assumed

Data integrity, licence terms, identifiers, image and label geometry, and available evaluation procedures will be documented before the protocol freeze.

6.2 Primary Analysis Boundary

- The primary analysis uses only the usable labelled training data.
- The 269 scans remaining after the development split each receive exactly one held-out prediction from the corresponding fold.
- Official ATM'22 validation and hidden test sets are not included in cross-validation.
- If the official long-term evaluation procedure is accessible and permits a submission after the protocol freeze, one locked external evaluation may be run as a separate extension.

6.3 Development Set

Size	30 scans
Selection	Stratified random draw with a recorded random seed
Stratification variables	Available source or site information; original through-plane spacing
Selection constraint	Scan selection does not depend on model performance
Pilot training pool	20 scans
Pilot validation pool	10 scans
Use	Preprocessing assessment, metric verification, working spacing, computational feasibility, training-budget rule, approximate diameter thresholds
After protocol freeze	All 30 development scans are included as training-only cases in each cross-validation model; never part of held-out or inner-validation sets

> All development-set decisions are completed before any cross-validation held-out score is inspected.

6.4 Five-Fold Cross-Validation

The remaining 269 scans are assigned to five fixed, non-overlapping held-out folds: four of approximately 54 scans and one of approximately 53 under the expected usable count. Exact assignments and sizes are recorded after exclusions.

Grouping and stratification

- Where reliable patient identifiers exist, folds are grouped by patient.
- Where metadata permit, a fixed allocation procedure also balances source or site, original through-plane spacing and scan size across folds.

- No assignment depends on model performance.
- Where identifiers are absent or unreliable, duplicate and near-duplicate screening uses image headers and image-content checks where feasible. Residual uncertainty about patient-level independence is stated as a limitation.

Per-fold procedure

Step	Action
1	The held-out fold is set aside and remains unavailable for tuning, checkpoint selection and stopping.
2	Approximately 10% of the remaining cross-validation cases are assigned to a fold-specific inner-validation set using a recorded seed.
3	The same inner-validation scans are used for the 2D, 2.5D and 3D models in that fold.
4	The remaining cross-validation cases, together with all 30 development cases, form the training data.
5	Each model is trained independently and evaluated once on the held-out fold using its selected checkpoint.

The approximate training count per fold after inner validation and inclusion of the development cases is 223–224 scans, subject to final exclusions and grouping constraints.

6.5 Annotation and Reference Limitations

ATM'22 reports that its airway structures were delineated by three experienced radiologists. The released annotations are the reference target, but the mask is not assumed to be perfect anatomical truth at every voxel. Image resolution, partial-volume effects, annotation conventions and the limits of visual assessment can affect which peripheral branches are included. All recall and precision measurements are interpreted as agreement with the released reference, and precision is reported cautiously, particularly where a prediction extends into regions the reference may not delineate.

6.6 Data Quality Checks

Before training, all scans undergo predefined checks for:

- Unreadable or corrupted image volumes.
- Image and mask shape, orientation and affine mismatches.
- Invalid spacing or inconsistent geometry.
- Non-standard intensity representation.
- Duplicate or near-duplicate scans.
- Missing or unusable reference labels.

Exclusions are documented with case identifiers and reasons. No scan is excluded on the basis of a model's prediction quality.

7. Methods

7.1 Preprocessing

7.1.1 Orientation and geometry

- All scans are converted to a consistent anatomical orientation before axial slices or 3D patches are extracted.
- Original orientation, image affine and voxel spacing are preserved so predictions map back to the reference geometry.
- Original in-plane and through-plane spacing are recorded for every scan and summarised before the main experiment.

7.1.2 Working spacing

Candidates	Isotropic 1.0 mm and 0.75 mm
Selection rule	The candidate closest to the median original through-plane spacing of the development scans is considered first. It must also satisfy the round-trip metric criterion in Section 11.
Fallback	If the first candidate fails, the other is assessed. If neither satisfies the criterion, preprocessing and the round-trip test are reviewed before the protocol is frozen.
Constraint	No spacing is chosen after inspecting held-out model performance.
Physical meaning	The distance represented by one neighbouring slice equals the selected working spacing, and is reported to make the 2.5D context explicit.

7.1.3 Interpolation

- CT images: cubic spline interpolation, subject to confirmation of implementation and boundary behaviour during the pilot.
- Training masks: nearest-neighbour interpolation to avoid fractional labels.
- The chosen procedures are fixed in Appendix A.

7.1.4 Intensity processing

HU conversion	CT values converted to Hounsfield units where original metadata require rescaling.
Clipping	Planning window [-1000, 400] HU; final limits confirmed during the pilot.
Caching	Clipped and resampled CT volumes are cached without fold-specific normalisation.
Fold-specific statistics	Per fold, mean and standard deviation are computed from all valid voxels in the fold's training scans (including background air, excluding artificial padding). The calculation uses the cross-validation training cases and the 30 development training-only cases, but not that fold's inner-validation or held-out scans.
Normalisation	Z-score on the fly: x' = (x - mean) / SD. The same statistics are used for training, inner validation and held-out inference within the fold, so no held-out information enters the normalisation parameters.

7.1.5 Crops and lung regions

The core experiment uses no lung mask or externally generated lung crop, so results do not depend on an additional segmentation method. Predictions are generated in the working geometry and mapped back to the original reference geometry before evaluation.

7.1.6 Probability mapping

For evaluation, the full-volume probability map at working spacing is linearly resampled to the reference grid and thresholded at 0.5. This ordering is identical for all three models and is used in the round-trip test.

7.2 Model Architectures

All models use MONAI U-Net implementations with the same number of resolution levels, base channel width, normalisation and activation where dimensionality permits. Convolution dimensions differ according to input dimensionality.

> **Planning configuration**
> Five resolution levels, 16 initial feature maps, instance normalisation and LeakyReLU activation. Precise widths, convolutional units, strides and padding are recorded in Appendix A.

7.2.1 2D U-Net

Receives one axial CT slice, S(i), as a single-channel input and predicts a binary airway probability map for that slice.

7.2.2 2.5D U-Net

Receives three adjacent axial slices, [S(i-1), S(i), S(i+1)], and predicts only the airway probability map of the central slice, S(i).

Volume boundaries	Replicated edge slices
Shared configuration	2.5D and 2D share base architecture, crop size, loss, optimiser, update budget and inference-window policy.
Parameter count	The first convolution has more input channels in 2.5D, so the parameter count differs slightly. This is inherent to adding neighbouring slices and is reported.
Interpretation	The 2D versus 2.5D comparison estimates the incremental effect of adjacent-slice input context within this architecture.

7.2.3 Lightweight 3D U-Net

Same number of resolution levels and initial feature width, with 3D convolutions and patch-based training.

Patch size candidates	{96, 128, 160, 192} voxels per side
Selection rule	The largest candidate that fits the GPU with batch size 1 or 2, mixed precision and at least 10% memory headroom. For the selected patch, the largest feasible batch size among 1 and 2.
Gradient checkpointing	May be used as a feasibility measure if a candidate does not fit; its effect on training time is recorded.
Freezing	Final patch and batch size are fixed before the main runs.
Receptive field	The theoretical receptive field of the frozen architecture is calculated and reported with the patch size and effective spatial context. It is not a pass/fail condition because it may exceed every feasible patch.
Interpretation	A comparison of practical configurations. Patch size, receptive field, voxels per update, sampling units and memory differ, so observed differences cannot be attributed to dimensionality alone.

7.3 Training Protocol

The three models share one optimisation and loss framework. Differences needed to accommodate dimensionality and memory are recorded.

7.3.1 Optimisation

Setting	Value
Optimiser	AdamW
Initial learning rate	1e-3
Weight decay	1e-5
Learning-rate schedule	Cosine decay over the fixed update budget
Precision	Mixed precision; exact implementation recorded
Loss	Equally weighted soft Dice loss and binary cross-entropy with logits
Output	One airway logit per voxel; sigmoid for inference
Prediction threshold	Fixed at 0.5, not selected using held-out data

The Dice implementation, reduction and treatment of empty patches are frozen in Appendix A.

7.3.2 Training inputs

2D and 2.5D crop size	256 × 256 in-plane crops, with the same crop and padding rules
3D input	Cubic patches of the selected size
Normalisation consistency	Instance normalisation uses statistics from the current input. To avoid a crop-versus-full-slice discrepancy, 2D and 2.5D inference uses 256 × 256 sliding windows matching training.
3D inference windows	Equal to the training patch size
Constraint	No model is evaluated with a spatial window size different from the one on which it was trained.

7.3.3 Update-budget selection

A single common update budget, U, is used for all models in the main evaluation.

Candidate budgets	10,000, 20,000 and 40,000 optimiser updates
Pilot design	Each candidate is a separate pilot run with its own cosine schedule: 3 models × 3 budgets = 9 jobs.
Pilot data	20 development scans for training; remaining 10 for validation
Selection rule	The smallest candidate among 10,000 and 20,000 for which no larger tested budget yields an improvement of 1 absolute percentage point or more for any of the three models.
Fallback	If no common candidate is found, 40,000 updates are used and described as a fixed-budget choice, not a demonstrated convergence point.
Tolerance interpretation	The 1-point tolerance is a pilot budget-selection rule only. It is not a clinical threshold, acceptable-loss margin or evidence of equivalence.
Constraint	Completed on the development set before the freeze; not modified in response to held-out results or inner-validation performance from another fold.
Disclosure	Main-run training curves are retained and reported. If a model is still improving substantially at the final update, this is disclosed as a limitation.

7.3.4 Checkpoint selection

Evaluation points	Each run is evaluated on its inner-validation set at K evenly spaced update points, chosen in the pilot and applied identically to all models. Planning value: ten; five only if the pilot shows ten is computationally prohibitive.
Score	At each point, BD and TD are computed per scan and averaged across validation scans with equal weights. The mean of these two means is the primary checkpoint-selection score.
Tie-breaker	Scan-mean Dice, used only to break ties.
Empty prediction	BD = 0 and TD = 0. Undefined values follow the frozen evaluator rules and are reported.
Recording	Selected checkpoint index and validation score are recorded for every run.

7.3.5 Sampling

Scan selection	For each training example, a training scan is selected uniformly.
Slice or patch selection	A slice or patch is then selected from that scan.
Foreground fraction	A fixed fraction, provisionally 0.5, of samples contain at least one positive airway voxel; the rest are drawn without requiring foreground. The operational definition is fixed separately for 2D slices, 2.5D central slices and 3D patches.
Exposure reporting	Slices and patches are not equivalent sampling units. The study reports foreground fraction per batch, distinct scans sampled, total input voxels processed and total examples seen. Training exposure is not described as equal across dimensionalities.

7.3.6 Data augmentation

Augmentation is matched as closely as dimensionality permits. Exact ranges and probabilities are recorded in Appendix A.

Augmentation	Range	Probability
In-plane rotation	Up to 15 degrees	0.2
In-plane scaling	0.9 to 1.1	0.2
Intensity scaling and shifting	Provisionally limited to 10%	0.2
Additive Gaussian noise	Provisionally SD 0.05 in normalised intensity units	0.2

- Spatial transformations are applied consistently to images and labels.
- The same transformation is applied to every slice in a 2.5D input window.
- Corresponding in-plane transformations are used for 3D patches.
- Out-of-plane rotation is not used.

7.3.7 Random seeds

Primary seed	[42] for each model and fold
Recorded seeds	Python, NumPy, PyTorch, CUDA and data-loader seeds are set and documented
Seed-sensitivity analysis	Two additional seeds, [17] and [123], on one fold selected before any held-out result is inspected
Analysis constraint	Additional runs are analysed separately and not pooled as independent patients

7.4 Inference

7.4.1 2D and 2.5D inference

- MONAI sliding-window inference over axial slices of the full volume.
- In-plane window 256 × 256, 50% overlap, Gaussian blending.
- Slices smaller than the window are padded using the frozen padding convention.
- The 2.5D model receives the neighbouring slices associated with each central-slice location.
- Central-slice probabilities are combined by the fixed blending policy into a full-volume probability map.
- Window batching is selected in the pilot and held constant across 2D and 2.5D inference.

7.4.2 3D inference

- Sliding-window inference with a window equal to the selected training patch size.
- 50% overlap with Gaussian blending.
- Inference precision and window batching are frozen during the pilot.

7.4.3 Evaluation

Raw probability maps are evaluated using the prescribed mapping and thresholding procedure. No architecture-specific post-processing is used in the primary comparison. Any post-processing, refinement or connectivity repair is a separate extension and reported separately.

8. Evaluation Metrics

8.1 Primary Outcomes

Metric	Definition	Implementation
Branch detection rate (BD)	Proportion of reference airway branches detected by the prediction	Official ATM'22 evaluator
Tree length detected rate (TD)	Fraction of the reference airway tree length detected by the prediction	Official ATM'22 evaluator

Implementation requirements

- Both measures are computed per scan using the official ATM'22 evaluation code wherever accessible.
- The version or commit, input requirements, matching rules, connected-component handling and aggregation procedure are recorded before the main runs.
- The official definitions determine whether all predicted components or a designated component are considered. The benchmark's metric implementation is not silently replaced by a different convention.
- If an independent implementation is necessary, it is validated against the official definitions and available test cases before the freeze.
- One BD and one TD score per held-out scan per model. Overall means weight each scan equally rather than pooling branches or centerline elements across scans.

8.2 Secondary Outcomes

8.2.1 Dice and precision

Metric	Definition	Reporting
Dice	Dice similarity coefficient	Per scan; equal scan weights
Precision	Official ATM'22 definition if supported	Exploratory secondary outcome

8.2.2 Diameter-stratified recall

Estimates recovery of the available reference airway skeleton across approximate diameter groups. It is not a direct anatomical diameter measurement.

Step	Procedure
1. Reference skeleton	Three-dimensional skeleton of the reference airway mask using a documented skeletonisation implementation.
2. Approximate local diameter	At each reference skeleton voxel, twice the physical Euclidean distance-transform value. Sensitive to irregular boundaries, bifurcations, voxelisation and partial-volume effects.
3. Diameter bins	Three bins from tertiles of the pooled reference skeleton diameter distribution in the 30 development scans. Thresholds in physical units, frozen before held-out evaluation. Pilot check: each bin must contain at least [200] skeleton voxels in at least [90%] of development scans; otherwise three fixed physical intervals are defined and approved before the main evaluation. The same thresholds apply to every held-out scan.
4. Voxel recovery	A reference skeleton voxel is recovered if at least one predicted positive voxel lies within a fixed Euclidean distance of [1.0 mm, or the frozen equivalent] in physical space.
5. Per-scan recall	Computed per diameter bin and scan. A scan with fewer than [20] reference skeleton voxels in a bin has no valid score for that bin; excluded scans are reported for every bin.

8.2.3 Branch-point sensitivity analysis

- Branch points identified from the reference skeleton graph using 26-connectivity.

- Skeleton spurs shorter than [2 mm] pruned using a documented procedure before branch-point detection.
- Adjacent branch-point voxels of the same junction treated as one junction region.
- Diameter-stratified recall repeated after excluding skeleton voxels within [3 mm] of a branch point.
- A second sensitivity analysis shifts the frozen diameter thresholds by 10% and repeats the bin-level assessment.

8.2.4 Pre-specified RQ4 analysis

Eligibility	Only scans with valid recall in both the smallest and the largest diameter bins.
Contrast per scan	(2.5D recall in smallest bin minus 2D recall in smallest bin) minus (2.5D recall in largest bin minus 2D recall in largest bin)
Reporting	Mean of paired per-scan contrasts with 95% fold-stratified paired bootstrap interval. Eligible scans overall and per fold are reported.
Fallback	If fewer than [50%] of held-out scans are eligible, RQ4 is reported descriptively without an inferential conclusion.
Constraint	All other diameter analyses are exploratory and do not alter primary decisions.

8.2.5 Fragmentation

Metric	Definition
Connected components	Number of 26-connected components in the thresholded raw prediction
Outside-largest proportion	Proportion of predicted airway voxels outside the largest component
Empty prediction	Counted as zero components and flagged separately

Fragmentation is descriptive only. A prediction recovering only the central airway may have few disconnected components despite missing much of the peripheral tree, so it is not interpreted independently as evidence of anatomical continuity.

8.3 Computational Outcomes

Training memory, training time, inference latency and inference memory are measured as described in Section 10. Parameters, model size and operation counts are reported descriptively. No single computational metric is treated as a complete measure of efficiency.

9. Statistical Analysis Plan

The analysis plan will be reviewed by the advisor and, where possible, a statistician before the protocol freeze.

9.1 Unit of Analysis

- The held-out scan is the unit of analysis.
- Each scan contributes one primary prediction from its assigned fold and selected checkpoint.
- For each pair of models, BD and TD differences are calculated within each held-out scan before aggregation.
- The primary effect estimate is the mean paired difference across eligible held-out scans, with equal weighting per scan.

9.2 Primary Interval Procedure

Method	Fold-stratified paired percentile bootstrap
Resamples	10,000
Random seed	Pre-specified
Resampling unit	Within each of the five held-out folds, scans are resampled with replacement, preserving the original fold size
Pairing	Paired predictions for each resampled scan stay together
Statistic	Equally scan-weighted mean paired difference across resampled scans
Interpretation	Intervals represent scan-sampling uncertainty conditional on the five fitted fold models. Because folds have overlapping training sets and each fold has one primary initialisation per model, they are not interpreted as fully capturing training-set or retraining variability.

9.3 H1: 2.5D Superiority Over 2D

Metrics	BD and TD separately
Interval	Two-sided 97.5% bootstrap interval for the mean difference (2.5D minus 2D)
Metric-specific support	The entire interval lies above zero
General claim	Both primary metrics must meet their criteria. If only one does, the conclusion is metric-specific and partial.
Reported values	Effect estimate, interval endpoints, median paired difference, proportion of scans favouring each model

9.4 RQ2 and Conditional H2

Always reported	Two-sided 95% paired bootstrap intervals for (2.5D minus 3D) for both BD and TD
If a margin is approved	The lower endpoint of the two-sided 95% interval must exceed minus delta for both metrics
If no margin is approved	No non-inferiority claim. The models are compared descriptively through paired effect estimates and intervals.

9.5 Supporting Analyses

- Median paired differences.
- Proportion of held-out scans on which each model has the higher score.
- Mean paired differences in each of the five folds.
- Leave-one-fold-out sensitivity estimates showing how the overall mean difference changes when each fold is omitted.
- Paired Wilcoxon signed-rank tests for BD and TD, with Holm-adjusted p-values.

These do not replace or modify the primary interval-based decisions. Any important disagreement between primary and supporting analyses is discussed explicitly.

9.6 Non-Inferiority Margin

If tested, delta is expressed in absolute percentage points and represents the maximum acceptable loss in BD and TD. The justification document must include:

1. The practical interpretation of the allowed difference.
2. The observed seed sensitivity on the pre-selected fold.

3. Performance differences reported in ATM'22 benchmark results, acknowledging the limits of comparing whole pipelines.
4. The advisor's assessment of the acceptable loss for the intended research use.
5. The confidence-interval procedure used in the decision.

> Seed variability from one fold and differences among challenge submissions are limited evidence for an acceptable-loss margin. They are contextual evidence, not sufficient justification alone. If no adequately defensible delta is agreed, RQ2 remains estimation-only.

9.7 Secondary and Exploratory Analyses

Dice, precision, diameter-stratified recall, fragmentation, slice-spacing analyses and cost outcomes are reported with descriptive summaries and intervals where practical. They are labelled exploratory or secondary, and no confirmatory claims rest on unadjusted exploratory comparisons. If BD and TD disagree, both are reported and no unqualified statement of improved overall completeness is made.

9.8 Random-Seed Sensitivity

- Two additional seeds are run on the pre-selected fold.
- The range and spread of BD and TD across the three seeds are reported separately for each model.
- These runs are not treated as independent patients, pooled with primary held-out scans, or used as general estimates of retraining variability.

10. Computational Cost Measurement

Completeness is reported alongside cost. No architecture is described as universally most efficient based on one measurement.

10.1 Primary Computational Outcomes

10.1.1 Peak training memory

Measurement	Peak allocated and peak reserved GPU memory from PyTorch CUDA memory statistics
Recorded configuration	Batch size or patch size, mixed-precision setting, gradient checkpointing setting, input voxels per sample
Interpretation	Memory results are interpreted in relation to each model's specified configuration

10.1.2 Training time

Wall-clock time is separated into optimiser update time, validation time, one-time preprocessing time, and total training and validation time for the run. Optimiser updates, samples, input voxels processed, distinct training scans seen and GPU-hours are recorded. All timings use the same machine and consistent procedures.

10.2 Secondary Computational Outcomes

10.2.1 Inference latency

Level	Definition
Model latency	Time from a preprocessed volume held in CPU memory to a complete full-volume probability map at working spacing. Includes GPU transfer, window batching, network execution, blending and reassembly.
Pipeline latency	Model latency plus resampling to working spacing and mapping back to the reference geometry.
Excluded	Disk loading and metric computation, reported separately where useful.

Timing protocol

- Same pre-selected scans and machine settings for each model.
- Preferred timing set: every scan in a held-out fold selected before any results are examined. If infeasible: [10] scans chosen by a recorded random seed and stratified by volume size and original spacing.
- Two warm-up passes discarded; each scan timed over five repetitions, with GPU synchronisation around each timed section.
- Per-scan median latency and interquartile range reported.
- Inference window count per scan reported for all three models.

10.2.2 Inference memory

Peak allocated and reserved inference memory for the complete inference configuration, including the selected sliding-window batching.

10.3 Descriptive Computational Outcomes

Reported where practical: trainable parameter count, model file size, floating-point operations per forward pass, approximate operations per full scan, and throughput in scans per hour. Operation counts use a documented method and the exact input configuration; if full-scan counts are estimated rather than measured, that is stated.

10.4 Hardware and Reproducibility

All experiments use one fixed machine. The configuration is recorded before the protocol freeze, and the GPU memory reserve, training configurations and inference settings are included in the released documentation where possible.

Component	Specification
GPU	[model and VRAM]
CPU	[model]
System RAM	[amount]
Operating system	[version]
CUDA	[version]
PyTorch	[version]
MONAI	[version]
Other libraries	[versions]

11. Pilot, Protocol Freeze and Go/No-Go Criteria

11.1 Pilot Design

The pilot uses only the 30 development scans and the fixed 20/10 split from Section 6.3. No held-out fold is examined. It assesses:

- Dataset integrity and reproducibility of preprocessing.
- Metric-code correctness.
- Working-spacing and round-trip suitability.
- 3D patch and batch feasibility.
- Inference-window implementation.
- Candidate training budgets.
- Diameter-bin adequacy.

- Full workload and compute feasibility.

All substantive choices are recorded in Appendix A and the statistical analysis plan.

11.2 Planned Training-Job Count

Category	Jobs
Pilot update-budget search (3 models × 3 budgets)	9
Core cross-validation (3 models × 5 folds)	15
Conditional seed sensitivity (3 models × 2 additional seeds)	6
Total planned full training jobs (before technical-failure reruns)	**30**

A further three jobs may be needed if a documented technical issue requires one fold's three models to be repeated. Additional 3D memory-feasibility tests and configuration dry-runs are recorded separately and included in the GPU-hour estimate. The core experiment comprises 9 pilot and 15 cross-validation jobs; the six seed-sensitivity jobs depend on available compute.

11.3 Compute Feasibility

- The pilot measures GPU-hours per run type, including validation passes.
- The total estimate includes the pilot search, all 15 core runs, preprocessing, inference, dry-runs and a contingency for failures.
- A reserve of [20%] of the estimated required compute is maintained.
- Available GPU-hours and expected workload are assessed before the freeze.
- If the full workload is infeasible, the optional seed-sensitivity runs are removed before the core experiment is changed.
- If the core three-model comparison is infeasible, the study scope is revised before main training begins.

11.4 Go Criteria

The protocol may be frozen only when all of the following are satisfied:

#	Criterion
1	**Evaluation code.** The official ATM'22 code runs successfully, or an alternative implementation matches the official definitions on at least [10] development scans within [0.1 percentage point], subject to agreement on the test procedure.
2	**Round-trip fidelity.** Round-tripping reference masks through the chosen working spacing, with the same interpolation and thresholding path as the final predictions, yields mean BD and TD of at least [98%] relative to the original masks across the development set. If the first spacing fails, the other is tested; if both fail, preprocessing is reviewed.
3	**Geometry.** All development-scan prediction grids map back exactly to their reference grids in shape, spacing, origin and direction.
4	**Input policy.** Training and inference window sizes and padding procedures are tested and implemented. No model is evaluated on a different spatial window size from training.
5	**Training stability.** All three models complete the pilot without numerical divergence or unresolved software errors. A fixed-budget designation is used wherever the budget search fails to establish the defined pilot plateau.
6	**Inference.** Each model completes full-volume inference with its frozen window configuration on the available machine.
7	**Diameter analysis.** The proposed bins satisfy the adequacy criteria, or approved fixed physical bins are adopted.
8	**Compute budget.** The estimated core workload, configuration tests and contingency fit within the available compute budget and approved reserve.
9	**Protocol approval.** Appendix A, the statistical analysis plan and the complete protocol are version-stamped and approved in writing by the advisor before the first primary run.

> No tuning based on cross-validation held-out scores is permitted. Once frozen, the update budget, architecture, working spacing, evaluation procedure and statistical decision rules remain fixed.

11.5 3D Feasibility Gate

The 3D arm is feasible if it:

- Trains using at least the minimum candidate patch of 96 × 96 × 96 voxels.
- Uses batch size one.
- Uses mixed precision and gradient checkpointing where necessary.
- Completes the selected update budget within the approved compute estimate.

If the 3D arm fails this gate, the study does not silently proceed as a two-model experiment under the original title. The advisor approves a documented revision of the title, primary aims and hypotheses into a 2D versus 2.5D study before any main run begins.

11.6 Protocol Amendments

- Once frozen, no methodological changes are made in response to observed cross-validation performance.
- A correction for a software defect or numerical failure is documented with its reason and impact.
- Any substantive amendment after the primary experiment starts is versioned and identifies which runs require repetition.
- If any held-out results have already been examined, results from an amended analysis are clearly distinguished from the original pre-specified analysis.

12. Risks and Limitations

#	Risk	Mitigation and interpretation
12.1	**3D memory and training time.** A practical 3D model may remain infeasible on the available GPU.	Mixed precision, batch size one, smaller patches and gradient checkpointing mitigate memory but cannot guarantee feasibility. If the minimum viable 3D configuration fails, the scope is revised formally.
12.2	**Limited 3D context.** The selected patch may be much smaller than patches used by earlier low-memory airway networks.	Smaller patches restrict context and may limit completeness. The 3D result is the performance of the specified practical model, not a general upper bound for 3D segmentation.
12.3	**Unequal computational exposure.** The networks process different sampling units with different input sizes and costs.	Equal optimiser updates do not imply equal computational work or image exposure. These differences are reported explicitly.
12.4	**Inference-window overhead.** Fixed 256 × 256 windows for 2D and 2.5D may increase inference time versus full-slice inference.	Required for consistent input dimensions with training under instance normalisation. Window count and measured latency are reported.
12.5	**Interpolated neighbouring-slice context.** Resampling may introduce interpolated slices, especially for thick original through-plane spacing.	A three-slice window does not guarantee three independent observations. Original spacing is recorded, and a pre-specified sensitivity analysis examines whether the 2D-to-2.5D difference changes after excluding scans above the chosen spacing threshold.
12.6	**Pilot-based budget selection.** The update budget is chosen using a small development training pool.	It may not predict the optimal duration with 223–224 training scans per run. The procedure is frozen before cross-validation, and learning-curve limitations are reported rather than corrected through outcome-driven adjustments.
12.7	**Limited retraining inference.** Five folds with one initialisation per model per fold say little about seed and training-set variability.	The seed check is a single-fold sensitivity assessment unless more resources are available. Results are comparisons conditional on the fitted models.

#	Risk	Mitigation and interpretation
12.8	**Reference-label limitations.** Released annotations come from experienced radiologists.	Reference-based metrics cannot establish absolute anatomical completeness. Small-airway visibility, annotation conventions and resolution remain limitations.
12.9	**Single-benchmark evaluation.** One established binary airway-segmentation benchmark.	Results do not automatically generalise to other acquisition protocols, scanners, populations or disease distributions.
12.10	**No assumed winner or clinical benefit.** The study may find no meaningful difference, a 3D advantage or a benefit from neighbouring slices.	All outcomes are reported as observed. Segmentation performance alone does not establish clinical utility.

13. Scope and Extensions

13.1 Core Experiment

Dataset	One: ATM'22
Models	Three: 2D, three-slice 2.5D and lightweight patch-based 3D U-Nets
Validation	Five-fold cross-validation
Primary outcomes	Two: BD and TD
Secondary metrics	Dice, precision, approximate diameter-stratified recall and fragmentation
Statistics	Pre-specified statistical analysis
Cost	Training and inference cost profiling
Reproducibility	Documented and reproducible implementation

13.2 Optional Extensions

Considered only after completion of the core experiment:

- A larger 3D patch experiment on one fixed fold to investigate sensitivity to patch size.
- A five-slice or strided-slice 2.5D model.
- A clDice-style or topology-aware loss and simple post-processing.
- A locked submission to the official ATM'22 long-term evaluation, if access and submission rules permit.
- External evaluation using AeroPath, reported separately because of contrast and pathology differences.
- A lung-crop experiment using one documented lung-segmentation tool.
- An nnU-Net reference experiment.
- A separate memory-constrained 3D experiment, with the constraint and changes documented.
- A separate evaluation on ATM'26, if suitable data access and evaluation conditions are available.

> No extension replaces the core experiment or is used to modify its primary conclusions retrospectively.

14. Work Plan

The planned duration is 32 weeks, subject to confirmation of available hardware and the pilot compute estimate.

Weeks	Phase	Activities
1–3	Registration and initial review	• Complete ATM'22 registration and verify data-use requirements • Download and inspect labelled scans; confirm usable case count, identifiers, geometry and hardware specifications • Complete focused literature review including recent airway studies and ATM'26; record search dates, strings and inclusion criteria
4–7	Data preparation and evaluation	• Implement geometry-preserving preprocessing and resampling • Verify image, mask and orientation consistency • Implement fixed development and cross-validation partitions • Obtain and test official ATM'22 evaluation code • Implement sliding-window inference for all three models • Validate probability-mapping and round-trip evaluation
8–11	Pilot and protocol freeze	• Three-model, three-budget pilot search • Measure 3D memory feasibility and full-volume inference costs • Finalise working spacing and diameter-analysis procedure • Complete Appendix A and statistical plan; obtain written advisor approval; freeze protocol by end of week 11
12–21	Main experiment	• Run 15 primary cross-validation jobs • Record training time, memory, GPU-hours and exposure; save and document selected checkpoints • Run optional seed-sensitivity jobs only if compute permits • No changes to the frozen protocol based on held-out results
22–27	Analysis and interpretation	• Compute paired BD and TD comparisons with pre-specified bootstrap intervals and supporting analyses • Conduct diameter-stratified analysis and approved sensitivity checks • Examine prediction failures; generate qualitative visualisations • Complete any feasible extensions without modifying primary results
28–32	Writing and dissemination	• Update literature search and record final date • Prepare manuscript with pre-specified methods and complete results • Verify references, tables, figures and statistical calculations • Release permissible code, configuration, folds and evaluation scripts • Select a publication venue appropriate to completed results

Literature notes, experiment logs and manuscript preparation continue throughout the project.

15. Expected Outputs and Publication Direction

15.1 Expected Outputs

- A documented implementation of 2D, 2.5D and lightweight 3D U-Nets for pulmonary airway segmentation.
- A paired comparison of BD and TD on the ATM'22 held-out cross-validation scans.
- Estimates and intervals for the performance difference between neighbouring-slice 2.5D and 2D.
- An estimate of the difference between 2.5D and lightweight 3D, with conditional non-inferiority assessment only if a justified margin is approved.
- A pre-specified approximate diameter-stratified recall analysis.
- A completeness-versus-cost profile: training memory, training time, inference latency and other computational measurements.
- A reproducible pipeline with documented preprocessing, folds, seeds, checkpoints, software versions and metric scripts.

Release of source code, fold definitions and evaluation scripts complies with the ATM'22 registration and data-use terms. CT scans and masks are not redistributed.

15.2 Publication Direction

Publication is not guaranteed. The contribution is an empirical, controlled comparison rather than a new architecture. A paper based only on overall Dice, or on confirming that one dimensionality outperforms another, could be considered incremental. The most informative results would establish:

- The size and direction of the completeness difference between 2D and 2.5D.
- Whether the incremental benefit of neighbouring slices changes between the smallest and largest approximate reference-diameter groups.
- How observed completeness differences relate to measured training and inference costs.
- The extent to which conclusions depend on the specified lightweight 3D patch configuration.

Workshop papers and short research articles may be appropriate, depending on the strength and completeness of the results. Potential venues include relevant MIDL or MICCAI workshops and journals interested in reproducible medical-image analysis. The venue is selected after results and limitations are known. No claim of clinical effectiveness is made from this benchmark study alone.

Appendix A: Protocol Freeze Table

All operational values must be completed and version-stamped before the primary runs.

A.1 Dataset and Partitioning

Parameter	Value
Usable ATM'22 training cases	[confirmed count]
Development scans	[30]
Pilot training/validation division	[20/10]
Cross-validation folds	[five fixed folds]
Group and stratification procedure	[final method]
Random seeds	[recorded values]

A.2 Preprocessing

Parameter	Value
Canonical orientation	[confirmed convention]
Working voxel spacing	[1.0 mm or 0.75 mm isotropic, selected by pre-specified rule]
Image interpolation	[cubic spline]
Mask interpolation	[nearest neighbour]
Image clipping	[-1000 to 400 HU, subject to confirmation]
Normalisation	[fold-specific training-scan pooled mean and standard deviation]
Padding value	[normalised equivalent of lower clipped HU bound]
Prediction mapping order	[linear probability resampling, then thresholding]
Original-spacing sensitivity threshold	[fixed value in mm]

A.3 Architecture

Parameter	Value
Network levels	[five]
Channel widths	[16, 32, 64, 128, 256, if confirmed]
Normalisation	[instance norm]
Activation	[LeakyReLU]
Convolution kernels	[confirmed configuration]
Downsampling and upsampling	[confirmed configuration]
Padding policy	[confirmed configuration]
Final output channels	[one]
Parameter counts	[computed values]

A.4 Training

Parameter	Value
Optimizer	AdamW
Learning rate	1e-3
Weight decay	1e-5
Learning-rate schedule	Cosine
Loss	0.5 Dice + 0.5 BCE with logits
Update budget U	[pilot-selected]
Checkpoint evaluations K	[10 or five, frozen]
Inner-validation fraction	[10% of available cross-validation training scans]
Foreground sampling fraction	[0.5]
2D/2.5D crop size	[256 × 256]
2D/2.5D batch size	[16, if confirmed]
3D patch size	[pilot-selected]
3D batch size	[1 or 2]
Gradient checkpointing	[enabled or disabled]
Augmentation values	[final numerical ranges and probabilities]
Prediction threshold	0.5
Primary training seed	42
Additional sensitivity seeds	17 and 123

A.5 Inference

Parameter	Value
2D/2.5D window	[256 × 256]
2D/2.5D overlap	[50%]
2D/2.5D blending	Gaussian
3D window	[selected training patch]
3D overlap	[50%]
3D blending	Gaussian
Inference precision	[fixed setting]
Window batch size	[pilot-selected]
Timing scans	[fixed list]

A.6 Evaluation

Parameter	Value
Official evaluator	[version or commit]
Metric inputs and connected-component treatment	[documented]
Bootstrap resamples	10,000
Bootstrap seed	[fixed value]
Primary confidence interval	[two-sided 97.5% for H1]
RQ2 estimation interval	[two-sided 95%]
Non-inferiority margin	[approved delta or estimation-only]
Diameter thresholds	[frozen physical values]
Matching tolerance	[fixed value in mm]
Minimum skeleton voxels per bin	[20 for per-scan recall validity]
Branch-point spur pruning	[2 mm, if confirmed]
Branch-point exclusion distance	[3 mm, if confirmed]
RQ4 minimum analysis-set retention	[50%, if approved]

A.7 Hardware and Software

Parameter	Value
GPU model and VRAM	[recorded]
CPU	[recorded]
RAM	[recorded]
Operating system	[recorded]
CUDA	[recorded]
PyTorch	[recorded]
MONAI	[recorded]
Other relevant packages	[recorded]

A.8 Approval

Item	Detail
Protocol version	[version]
Freeze date	[date]
Advisor approval	[written approval]
Statistician review	[completed or not available]

Appendix B: Literature Search and Screening Log

The focused literature review is completed before protocol freeze and updated before manuscript submission.

B.1 Record-Keeping Requirements

For each search, record:

- Actual search date.
- Database or source.
- Complete search string.
- Any date or language restrictions.
- Number of records retrieved.
- Number of records screened.
- Inclusion and exclusion decisions.
- Full references for included studies.
- Reason for excluding the closest potential direct comparisons.

B.2 Suggested Search Concepts

- "airway segmentation" AND "2D" AND "3D"
- "airway segmentation" AND "2.5D"
- "pulmonary airway" AND "neighbouring slices"
- "airway segmentation" AND "ATM'22" AND "U-Net"
- "airway segmentation" AND "branch detection" AND "tree length"
- "airway segmentation" AND "computational cost" OR "GPU memory"
- "airway segmentation" AND "topology" OR "skeleton" OR "continuity"
- "airway segmentation" AND "dimensionality comparison"

B.3 Screening Notes

- Publications from 2024 onward are specifically screened, including Skeleton2Mask, progressive curriculum learning, Cupido Loss, FDRAS and other relevant methods.
- Potentially direct comparisons are examined in full text and classified by task, input dimensionality, backbone, dataset, primary metrics and reported computational measurements.
- The gap and novelty statement is revised if the search identifies a sufficiently close prior comparison.

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